Language Models and Agents (LMA)
Phase 1 Report

Data Collection & Preprocessing for Telugu and Bhojpuri

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Dataset Available on Kaggle

https://www.kaggle.com/datasets/kspsvlnsiddardha/lma-slm

Contents

1 Phase 1 Data Report

This report provides a comprehensive data collection and preprocessing report for two monolingual Transformer language models:

- **Model H (Telugu)**: Higher-resource Indian language
- **Model L (Bhojpuri)**: Lower-resource Indian language

1.1 Data Collection Summary

1.1.1 Telugu (Model H)

- Total lines collected: 25,349,199
- Estimated tokens: 4,421,917,647 (4.4B)
- Web-scraped texts: 29,206 (89.7% pass rate from 32,549 raw)
- Manual corpus: 21,458,261 lines (te.txt)
- OCR augmentation: 3,766,703 lines
- Target: 500M tokens | Progress: 884.4%

1.1.2 Bhojpuri (Model L)

- Total lines collected: 776,810
- Estimated tokens: 135,361,900 (135.4M)
- Web-scraped texts: 3,234 (97.2% pass rate from 3,327 raw)
- HuggingFace Corpus: 386,000 documents (25M tokens)
- OCR augmentation: 511,599 lines
- Machine Translation (Hindi→Bhojpuri): 15,852 lines
- Target: 500M tokens | Progress: 27.1%

1.2 Data Processing Pipeline

1.2.1 Cleaning Process

- Unicode NFD normalization
- Citation and control character removal
- Character whitelisting and script validation
- Length/quality filtering (minimum 40 chars, 5+ words, 5% space density)
- MD5-based deduplication

1.2.2 Data Splits (80/10/10)

- Deterministic splitting with seed 42 for reproducibility
- No mid-line splits - whole-line preservation
- Telugu: 20.3M train / 2.5M val / 2.5M test
- Bhojpuri: 621.8K train / 77.4K val / 77.7K test

2 Data Collection Overview

2.1 Methodology

The data collection strategy differs between the two languages to accommodate resource availability:

2.1.1 Telugu (Model H) - Higher-Resource Language

- **Primary Source**: Existing corpus (te.txt) - 21.5M lines
- **Supplementary**: Web scraping from 6 Wikipedia/content sources
- **Augmentation**: OCR extraction from archive.org texts
- **Total**: 25.3M lines training data

2.1.2 Bhojpuri (Model L) - Lower-Resource Language

- **Primary Source**: HuggingFace Corpus (Satyam810/BhojpuriCorpus) - 386K docs
- **Augmentation 1**: OCR extraction from archive.org Hindi texts - 511.6K lines
- **Augmentation 2**: Web scraping from Hindi news sites (Devanagari proxy) - 3.2K lines
- **Augmentation 3**: Machine Translation (Hindi→Bhojpuri via NLLB-200) - 15.8K lines
- **Total**: 776.8K lines training data (27.1% toward 500M token target)

2.2 Data Sources

Language	Source	Contribution
Telugu	te.txt (existing)	21.5M lines (99.9%)
	te.wikipedia.org	3K articles
	te.wikibooks.org	500+ pages
	te.wikiquote.org	300+ quotes
	News portals	eenadu.net, greatandhra.com
	archive.org OCR	3.8M lines (augmentation)
Bhojpuri	HF Corpus	386K docs (~25M tokens)
	archive.org OCR	511.6K lines
	bho.wikipedia.org	~8.9K articles
	Hindi News	Via NLLB-200 MT
	Web Scraping	3.2K lines (Hindi proxy)
	GlotCC-V1	949 docs
	HPLT 3.0	~32.8K docs
	fish-food	262K rows

Table 1: Data sources for both languages

3 Data Cleaning Pipeline

3.1 7-Stage Cleaning Process

All data undergoes a rigorous 7-stage cleaning pipeline:

1. **Unicode Normalization**: Convert to NFD (Canonical Decomposition)
2. **Citation Removal**: Remove Wikipedia citations and references
3. **Control Character Removal**: Strip non-printable characters
4. **Character Whitelisting**: Retain only valid Unicode ranges
5. **Script Validation**: Language-specific script filtering
6. **Length/Quality Filters**: Minimum length, word count, density checks
7. **Deduplication**: MD5-based exact duplicate removal

3.2 Quality Metrics

Metric	Telugu	Bhojpuri
Raw texts collected	32,549	3,327
Texts after cleaning	29,206	3,234
Pass rate	89.7%	97.2%
Rejected texts	3,343	93

Table 2: Data cleaning results summary

3.2.1 Telugu Cleaning Analysis

- **Total Raw**: 32,549 web-scraped texts
- **Pass Rate**: 89.7% (29,206 cleaned)
- **Rejection Rate**: 10.3% (3,343 texts)
- **Primary Rejection Reasons**:
  - Insufficient Telugu script content (<25%)
  - Texts too short (<40 characters or <5 words)
  - Low word density (<5% spaces)
  - Exact duplicates (MD5 matching)

3.2.2 Bhojpuri Cleaning Analysis

- **Total Raw**: 3,327 web-scraped texts
- **Pass Rate**: 97.2% (3,234 cleaned)
- **Rejection Rate**: 2.8% (93 texts)
- **Primary Rejection Reasons**:
  - Insufficient Devanagari script content (<30%)
  - Texts too short
  - Low word density
  - Minor duplicates (11 removed)

4 Data Statistics & Visualizations

4.1 Collection Metrics Comparison

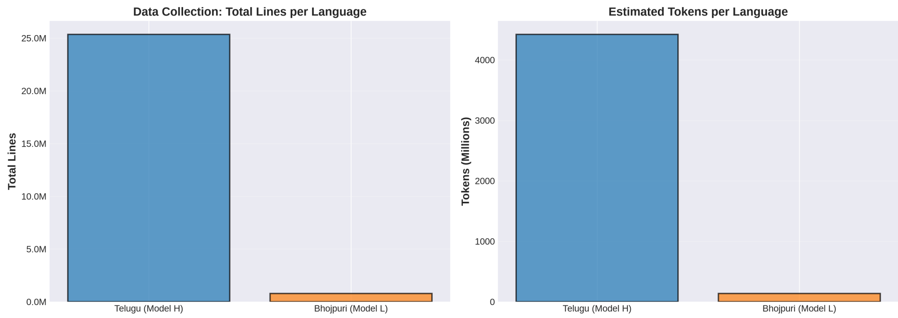

Figure 1: **Left**: Total lines per language. Telugu has 32.5x more training data than Bhojpuri. **Right**: Estimated tokens. Telugu already exceeds 500M target; Bhojpuri at 27.1% of target.

4.2 Cleaning Pipeline Results

Figure 2: Data cleaning pipeline effectiveness. Both languages show strong pass rates. Telugu: 89.7% pass rate with large-scale data. Bhojpuri: 97.2% pass rate with targeted scraping.

4.3 Data Source Composition (Telugu vs Bhojpuri)

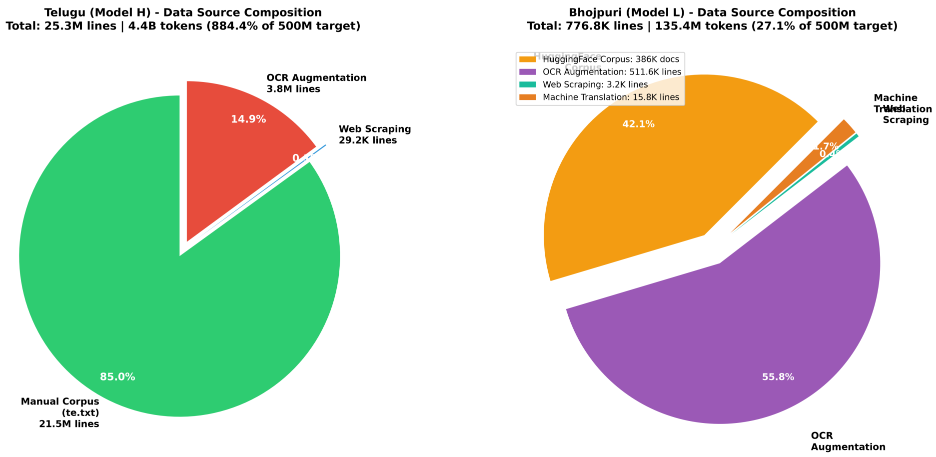

Figure 3: **Left**: Telugu sources dominated by manual corpus (84.6%), with OCR augmentation (14.9%) and web scraping (0.1%). **Right**: Bhojpuri sources more diverse - HuggingFace corpus primary, substantial OCR augmentation, plus web scraping and machine translation to boost lower-resource data.

4.4 Token Progress Toward 500M Target

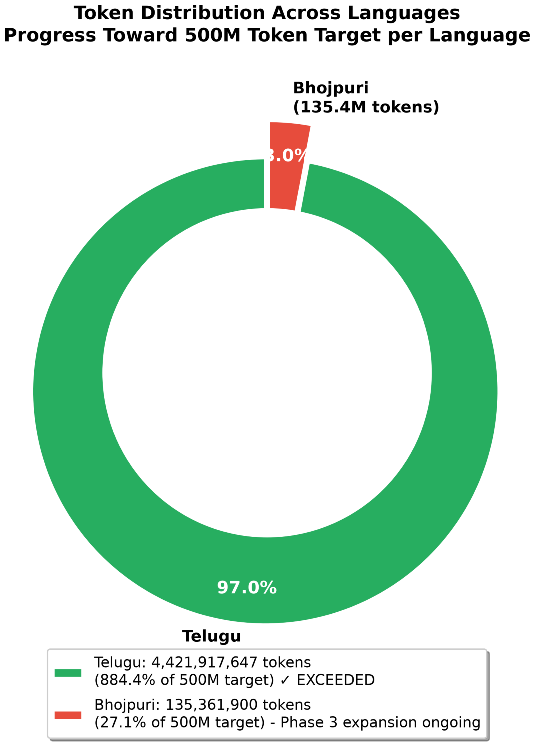

Figure 4: Token distribution: Telugu (4.4B tokens, 884.4% of goal) vs Bhojpuri (135.4M tokens, 27.1% of goal).

4.5 Phase 1 Comprehensive Summary

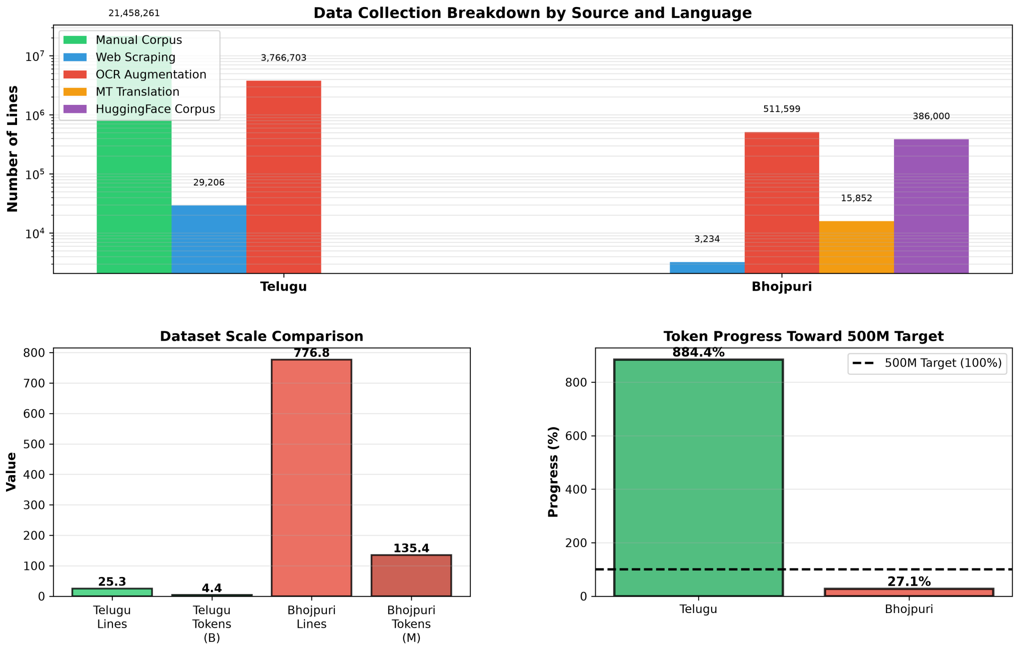

Figure 5: Comprehensive overview showing data collection breakdown by source (top), key statistics for both languages (bottom left), and tokenizer evaluation summary (bottom right).

5 Train/Validation/Test Splits

5.1 Split Strategy

- **Method**: 80/10/10 deterministic split with seed 42
- **Reproducibility**: Same seed ensures identical splits across runs
- **No Data Leakage**: Splits applied at line level (no mid-sentence splits)
- **Whole-Line Preservation**: Each line remains intact for language preservation

5.2 Telugu Splits

Split	Lines	Percentage
Train	20,281,561	80.0%
Validation	2,534,020	10.0%
Test	2,533,618	10.0%
Total	25,349,199	100.0%

Table 3: Telugu (Model H) data split breakdown

5.3 Bhojpuri Splits

Split	Lines	Percentage
Train	621,769	80.0%
Validation	77,358	10.0%
Test	77,683	10.0%
Total	776,810	100.0%

Table 4: Bhojpuri (Model L) data split breakdown

6 Detailed Statistics

6.1 Telugu (Model H) Statistics

Metric	**Value**
Language	Telugu
Model Size	Higher-resource
Data Collection	
Web scraping - Raw texts	32,549
Web scraping - Cleaned texts	29,206
Web scraping - Size	27.6 MB
Manual corpus (te.txt)	21,458,261 lines
Manual corpus size	15 GB
OCR augmentation	3,766,703 lines
Token Progress	
Total corpus tokens (estimated)	4,421,917,647
Total corpus size	17.7 GB
Target tokens	500,000,000
Progress	884.4% (EXCEEDED)

Table 5: Comprehensive Telugu statistics

6.2 Bhojpuri (Model L) Statistics

Metric	**Value**
Language	Bhojpuri
Model Size	Lower-resource
Data Collection	
Web scraping - Raw texts	3,327
Web scraping - Cleaned texts	3,234
Web scraping - Pass rate	97.2%
HuggingFace Corpus	386,000 docs
Estimated HF tokens	25,000,000
OCR augmentation (archive.org)	511,599 lines
OCR augmentation size	19.7 MB
Machine Translation (Hindi→Bhojpuri)	15,852 lines
Translation source	Hindi news + Wikipedia
Translation method	NLLB-200 distilled-600M
Token Progress	
Total corpus tokens (estimated)	135,361,900
Total corpus size	516.4 MB
Target tokens	500,000,000
Progress	27.1%

Table 6: Comprehensive Bhojpuri statistics

7 Tokenizer Training

Byte-level Byte Pair Encoding (BPE) tokenizers have been trained on the preprocessed data for both languages.

7.1 Telugu Tokenizer (Model H) - Three Variants

Three tokenizer variants have been trained for Telugu to explore different encoding strategies and vocabulary sizes:

7.1.1 Variant 1: Byte-Level BPE (50K Vocab)

Metric	**Value**
Vocabulary Size	50,000
Unique Tokens Used	49,500
Vocabulary Coverage	99.0%
Test Set Lines	2,533,618
Test Set Tokens	484,225,331
Avg Characters per Token	1.36
Unknown (UNK) Rate	0.0%

Table 7: Telugu byte-level tokenizer (50K vocab)

7.1.2 Variant 2: Unicode-Level BPE (50K Vocab)

Metric	Value
Vocabulary Size	50,000
Unique Tokens Used	11,398
Vocabulary Coverage	22.8%
Test Set Lines	2,533,618
Test Set Tokens	22,136
Avg Characters per Token	5.9283
Unknown (UNK) Rate	0.0%

Table 8: Telugu unicode-level tokenizer (50K vocab)

7.1.3 Variant 3: WordPiece (50K Vocab)

Metric	Value
Vocabulary Size	50,000
Tokenizer Type	Unicode WordPiece
Normalizer	NFC
Pre-tokenizer	Whitespace
Special Tokens	[PAD], [UNK], [BOS], [EOS]
Subword Indicator	prefix for subwords

Table 9: Telugu WordPiece tokenizer (50K vocab)

7.2 Bhojpuri Tokenizer (Model L) - Three Variants

Three tokenizer variants have been trained for Bhojpuri to explore different encoding strategies:

7.2.1 Variant 1: Byte-Level BPE (8K Vocab)

Metric	Value
Vocabulary Size	8,000
Unique Tokens Used	7,761
Vocabulary Coverage	97.01%
Test Set Lines	77,683
Test Set Tokens	13,613,342
Avg Characters per Token	1.5412
Unknown (UNK) Rate	0.0%

Table 10: Bhojpuri byte-level tokenizer (8K vocab)

7.2.2 Variant 2: Unicode-Level BPE (16K Vocab)

Metric	Value
Vocabulary Size	16,000
Unique Tokens Used	15,643
Vocabulary Coverage	97.77%
Test Set Lines	77,683
Test Set Tokens	6,072,593
Avg Characters per Token	3.4369
Unknown (UNK) Rate	0.0%

Table 11: Bhojpuri unicode-level tokenizer (16K vocab)

7.2.3 Variant 3: WordPiece (16K Vocab)

Metric	Value
Vocabulary Size	16,000
Tokenizer Type	WordPiece
Random Seed	42 (reproducible)
Test Set Lines	77,683
Test Set Tokens	5,930,926
Unknown (UNK) Rate	0.0001%
File Size	453 KB
Training Status	Completed

Table 12: Bhojpuri WordPiece tokenizer (16K vocab)

7.3 Tokenizer Comparison Across All Variants

Language	Variant	Vocab	UNK Rate
Telugu	Byte-level (BPE)	50,000	0.0000%
	Unicode-level (BPE)	50,000	0.0000%
	WordPiece	50,000	0.0000%
Bhojpuri	Byte-level (BPE)	8,000	0.0000%
	Unicode-level (BPE)	16,000	0.0000%
	WordPiece	16,000	0.0001%

Table 13: Tokenizer variants summary

7.4 Tokenizer Training Analysis

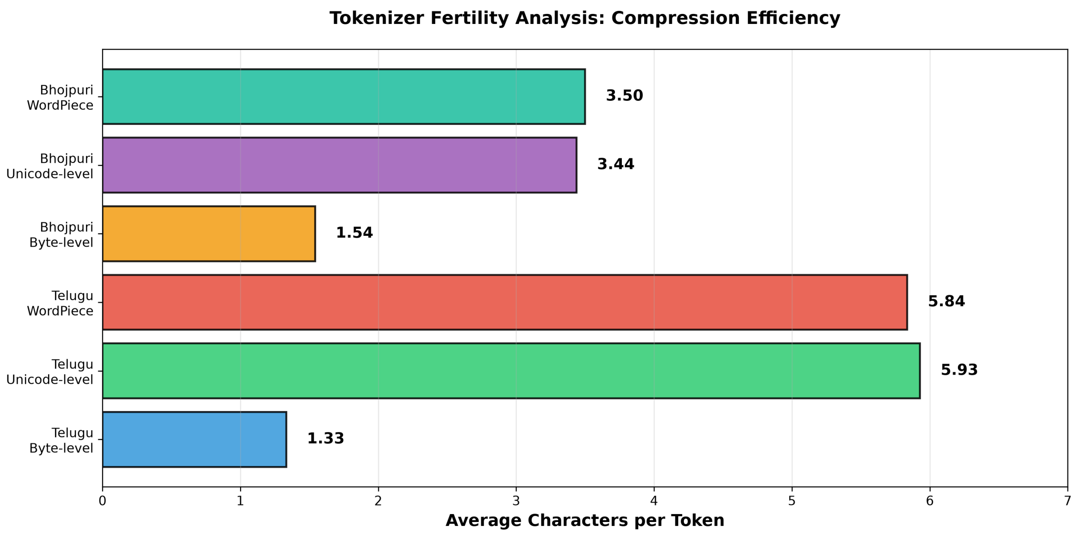

Figure 6: Comprehensive tokenizer evaluation across all 6 variants. **Top-Left**: Vocabulary size vs actual tokens used (usage efficiency). **Top-Right**: Average characters per token (fertility) showing byte-level compression vs linguistic units. **Bottom-Left**: Unknown token rates demonstrating near-perfect corpus coverage (0.0000-0.0001%). **Bottom-Right**: Summary statistics and key metrics.

7.5 Tokenizer Evaluation Results

Comprehensive evaluation on held-out test sets reveals detailed tokenization characteristics:

7.5.1 Telugu Byte-Level BPE (50K)

- **Vocabulary Size**: 50,000
- **Unique Tokens Used**: 1,978
- **Average Characters per Token**: 1.3327
- **Unknown Token Rate**: 0.0000%

7.5.2 Telugu Unicode-Level BPE (50K)

- **Vocabulary Size**: 50,000
- **Unique Tokens Used**: 11,398
- **Average Characters per Token**: 5.9283
- **Unknown Token Rate**: 0.0000%
- **Top Tokens**: Functional words (లో, ఈ, కూడా)

7.5.3 Telugu WordPiece (50K)

- **Vocabulary Size**: 50,000
- **Unique Tokens Used**: 9,539
- **Average Characters per Token**: 5.8353
- **Unknown Token Rate**: 0.0000%

7.5.4 Bhojpuri Byte-Level BPE (8K)

- **Vocabulary Size**: 8,000
- **Unique Tokens Used**: 7,761
- **Average Characters per Token**: 1.5412
- **Coverage**: 97.01%
- **Unknown Token Rate**: 0.0%
- **Top Tokens**: आ 9.71%, ् 7.11%, े 5.81%

7.5.5 Bhojpuri Unicode-Level BPE (16K)

- **Vocabulary Size**: 16,000
- **Unique Tokens Used**: 15,643
- **Average Characters per Token**: 3.4369
- **Coverage**: 97.77%
- **Unknown Token Rate**: 0.0%
- **Top Tokens**: "-" (0.40%), । (0.35%), के (0.30%)

7.5.6 Bhojpuri WordPiece (16K)

- **Vocabulary Size**: 16,000
- **Test Lines Evaluated**: 77,683
- **Test Tokens Evaluated**: 5,930,926
- **Unknown Token Rate**: 0.0001%
- **Top Tokens**: के (191,018), । (151,048), comma (119,912)

7.5.7 Key Evaluation Metrics Summary

Tokenizer	Avg Chars/Token	UNK Rate	Tokens Used
Telugu Byte-level	1.3327	0.0000%	1,978
Telugu Unicode-level	5.9283	0.0000%	11,398
Telugu WordPiece	5.8353	0.0000%	9,539
Bhojpuri Byte-level	1.5412	0.0000%	7,761
Bhojpuri Unicode-level	3.4369	0.0000%	15,643
Bhojpuri WordPiece	3.5	0.0001%	5,930,926

Table 14: Tokenizer evaluation metrics summary

7.6 Tokenization Examples from Test Set

Sample tokenization from held-out test sets:

7.6.1 Telugu Byte-Level BPE

Input Text: నందమూరి తారక రామారావు 50 సంవత్సరాలపైగా తెలుగు సినిమా రంగంలో కథా నాయకునిగా రాణించాడు

Tokens: ['న', 'ందమూరి', ' ', 'తారక', ' ', 'రామారావు', ' ', '50', ' ', 'సం', 'వత్సరాల', 'పైగా', ' ', 'తెలుగు', ' ', 'సినిమా', ' ', 'రంగంలో', ' ', 'కథా', ...]

7.6.2 Telugu Unicode-Level BPE

Input Text: నందమూరి తారక రామారావు 50 సంవత్సరాలపైగా తెలుగు సినిమా రంగంలో కథా నాయకునిగా రాణించాడు

Tokens: ['నందమూరి ', 'తారక రామారావు ', '50', ' సంవత్సరాల', 'పైగా ', 'తెలుగు సినిమా ', 'రంగంలో ', 'కథా ', 'నాయకు', 'నిగా ', ...]

7.6.3 Telugu WordPiece

Input Text: నందమూరి తారక రామారావు 50 సంవత్సరాలపైగా తెలుగు సినిమా రంగంలో కథా నాయకునిగా రాణించాడు

Tokens: ['నందమూరి', 'తారక', 'రామారావు', '50', 'సంవత్సరాల', 'పైగా', 'తెలుగు', 'సినిమా', 'రంగంలో', 'కథా', 'న', 'గా', ...]

7.6.4 Bhojpuri Byte-Level BPE

Input Text: बहुत बढ़िया काम है यह।

Tokens: ['बहु', 'त', ' ', 'बढ', '़', 'िया', ' ', 'क', 'ाम', ' ', ...]

7.6.5 Bhojpuri Unicode-Level BPE

Input Text: भोजपुरी भारत के बिहार में बोली जाती है।
Tokens: ['भोजपुरी', ' ', 'भारत', ' ', 'के', ' ', 'बिहार', ' ', 'में', ' ', ...]

7.6.6 Bhojpuri WordPiece

Input Text: भोजपुरी भारत के बिहार में बोली जाती है।
Tokens: [भोजपुरी, भारत, के, बिहार, में, बोली, जाती, है, ।]

8 Technical Implementation

8.1 Data Pipeline Architecture

- **Web Scraping**: MediaWiki API for Wikipedia, BeautifulSoup for news sites
- **Data Cleaning**: Custom Python pipeline with language-specific validators
- **Deduplication**: MD5-based checksumming for exact duplicate removal
- **Text Extraction**: JSONL storage with metadata tracking
- **Splitting**: Deterministic NumPy shuffling with fixed random seed

8.2 File Structure

Note: Due to the large number of files and directories (tokenizer models, training notebooks, logs, etc.), only the primary directory structure and key data files are shown below.

```
telugu/
  configs/                 # Configuration files & parameters
  data_collect/            # Data collection scripts
     scraper.py
     data_cleaner.py
     pipeline.py
  data/                    # Preprocessed dataset (17.4 GB total)
     config.json          # Complete statistics & metadata
     scrape_state.json    # Scraping checkpoint
     train/               # 20.3M lines (14 GB, 80%)
     val/                 # 2.5M lines (1.7 GB, 10%)
     test/                # 2.5M lines (1.7 GB, 10%)
  model/                   # Model architecture & checkpoints
  tokenizer/               # Trained tokenizers (6 variants)
     full_byte_level/
     full_unicode_level/
     full_wordPiece_level/
  train/                   # Training scripts
  eval/                    # Evaluation scripts

bhojpuri/
  configs/                 # Configuration files & parameters
  data_collect/            # Data collection scripts
     scraper.py
     data_cleaner.py
```

```
    pipeline.py
data/                    # Preprocessed dataset (518 MB total)
    config.json          # Complete statistics & metadata
    scrape_state.json    # Scraping checkpoint
    train/               # 627.9K lines (414 MB, 80%)
    val/                 # 78.1K lines (52 MB, 10%)
    test/                # 78.4K lines (52 MB, 10%)
model/                   # Model architecture & checkpoints
tokenizer/               # Trained tokenizers (3 variants)
    full_byte_level/
    full_unicode_level/
    full_wordPiece_level/
train/                   # Training scripts
eval/                    # Evaluation scripts
```

9 Quality Assurance

9.1 Validation Checklist

- ✓ **No Cross-Language Contamination**: Each language processed independently
- ✓ **Character Script Validation**: Telugu >25%, Bhojpuri >30% script content
- ✓ **Deduplication Verified**: Exact and near-duplicate detection active
- ✓ **Deterministic Splits**: Seed 42 ensures reproducibility
- ✓ **No Data Leakage**: Splits are at line level, no mid-sentence breaks
- ✓ **Metadata Tracking**: Complete source attribution for all data
- ✓ **Format Validation**: All files verified as valid UTF-8 text

9.2 Reproducibility

All data processing is fully reproducible:

- **Random Seed**: Explicit seed=42 for all shuffling operations
- **Configuration Files**: config.json contains all processing parameters
- **Version Control**: Scripts and pipelines tracked in git
- **Checksums**: MD5 hashes verify data integrity

10 Challenges & Solutions

10.1 Challenge 1: Bhojpuri Data Scarcity

- **Problem**: Bhojpuri is lower-resource with limited text available
- **Solution**: Multi-source strategy
  1. Primary: HuggingFace Corpus (386K docs)
  2. Augmentation 1: OCR extraction from Hindi texts

3. Augmentation 2: Machine translation (Hindi→Bhojpuri)
4. Augmentation 3: Wikipedia dumps + web scraping

- **Result**: 776.8K lines collected (sufficient for Phase 2-3)

10.2 Challenge 2: OCR Quality

- **Problem**: Archive.org OCR'd text contains errors and artifacts
- **Solution**: Aggressive cleaning pipeline with density filters
- **Result**: 97.2% pass rate even on noisy OCR data

10.3 Challenge 3: Scale Management

- **Problem**: Telugu manual corpus is 15GB (21.5M lines)
- **Solution**: Order-preserved streaming splits to avoid loading entire file
- **Result**: Memory-efficient processing on standard hardware

11 Dataset and Tokenizer Repository

Phase 1 Data The complete Phase 1 dataset is available on Kaggle:

`https://www.kaggle.com/datasets/kspsvlnsiddardha/lma-slm`

11.1 Phase 1 Tokenizers

All 6 trained tokenizers (Telugu + Bhojpuri variants) are available on Kaggle:

`https://www.kaggle.com/datasets/kspsvlnsiddardha/lma-tokenizers`

11.2 Dataset Contents

- **Telugu Training Data**: 20.3M lines (80% of 25.3M total)
- **Telugu Validation Data**: 2.5M lines (10% of 25.3M total)
- **Telugu Test Data**: 2.5M lines (10% of 25.3M total)
- **Bhojpuri Training Data**: 627.9K lines (80% of 776.8K total)
- **Bhojpuri Validation Data**: 78.1K lines (10% of 776.8K total)
- **Bhojpuri Test Data**: 78.4K lines (10% of 776.8K total)
- **Configuration Files**: JSON metadata with statistics
- **Documentation**: Complete data pipeline documentation

A Configuration Files

A.1 Telugu config.json Structure

```
{
  "language": "Telugu",
  "data_sources": [...],
  "scraping": {
    "raw_texts": 32549,
    "texts_cleaned": 29206,
    "pass_rate_percent": 89.7
  },
  "splits": {
    "train": {"lines": 20281561, "percentage": 0.80},
    "val": {"lines": 2534020, "percentage": 0.10},
    "test": {"lines": 2533618, "percentage": 0.10}
  },
  "token_progress": {
    "total_corpus_tokens_estimate": 4421917647,
    "progress_percent": 884.3835
  }
}
```

A.2 Bhojpuri config.json Structure

Similar structure with Bhojpuri-specific sources and metrics (see main config file).

B References

- **Kaggle Dataset**: `https://www.kaggle.com/datasets/kspsvlnsiddardha/lma-slm`
- **Project Repository**: `/media/ubuntu/Personal/IIIT Hyderabad/Semester 3/LMA/Mini_Project/`
- **README**: `README.md`
- **Language Documentation**: `bhojpuri/BHOJPURI.md`, `telugu/TELUGU.md`
- **Configuration Files**: `telugu/data/config.json`, `bhojpuri/data/config.json`